

Linear Algebra I: HW2 Partial Solution

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1.4-1. TFTFTF

1.4-5. YNNYYNYN

1.4-13. Given $s \in \text{span}(S_1)$, there exists $v_1, \dots, v_n \in S_1$ and $a_1, \dots, a_n \in F$ such that $s = a_1v_1 + \dots + a_nv_n$, but $v_1, \dots, v_n \in S_1 \subseteq S_2$, so $s \in \text{span}(S_2)$. If we further assume that $\text{span}(S_1) = V$, then $V = \text{span}(S_1) \subseteq \text{span}(S_2) \subseteq V$, which forces that $\text{span}(S_2) = V$.

1.4-14. Since $S_1, S_2 \subseteq S_1 \cup S_2$, we have $\text{span}(S_1), \text{span}(S_2) \subseteq \text{span}(S_1 \cup S_2)$, then by the minimality of sums, $\text{span}(S_1) + \text{span}(S_2) \subseteq \text{span}(S_1 \cup S_2)$. Conversely, since $S_1 \cup S_2 \subseteq \text{span}(S_1) + \text{span}(S_2)$, by the minimality of spanning set, $\text{span}(S_1 \cup S_2) \subseteq \text{span}(S_1) + \text{span}(S_2)$.

1.5-1. FTFFTT

1.5-2. DIIDDIDIID

1.5-15. ($\implies$) Suppose S is linearly dependent, if $u_1 = 0$, then we are done. Otherwise, $\{u_1\}$ is linearly independent. If $\{u_1, u_2\}$ is linearly dependent, then $u_2 \in \text{span}\{u_1\}$, and we are done (here $k = 1$). Otherwise, $\{u_1, u_2\}$ is linear independent. By doing so repeatedly, we can find a k , $1 \leq k < n$ such that $\{u_1, \dots, u_k\}$ is linearly independent and $\{u_1, \dots, u_{k+1}\}$ is linearly dependent. Consequently, $u_{k+1} \in \text{span}\{u_1, \dots, u_k\}$. ($\impliedby$) If $u_1 = 0$, then $0 \in S$ so S is linear dependent. If $u_{k+1} \in \text{span}\{u_1, \dots, u_k\}$ for some k , then there is a relation

$$u_{k+1} = a_1u_1 + a_2u_2 + \dots + a_ku_k, \text{ for some } a_1, \dots, a_k \in F \text{ not all zero}$$

Thus there is a nontrivial relation

$$a_1u_1 + a_2u_2 + \dots + a_ku_k - u_{k+1} = 0,$$

so S is linearly dependent.

1.5.20 Suppose on the contrary that there is a nontrivial relation

$$a_1e^{rt} + a_2e^{st} = 0.$$

Plug in $t = 0$, we have $a_1 + a_2 = 0$, thus $a_1 = -a_2$ and $a_1, a_2 \neq 0$. Plug in $t = 1$, we have $a_1(e^r - e^s) = 0$, so $e^r = e^s$, which implies $r = s$ and contradicts our assumption $r \neq s$.

3. The proof is mostly the same as that of 1.4.14. One needs to keep in mind that the sum of a family of vector subspaces collects vectors which are *finite sums* of vectors from the family.